

3.1 SPIN ECHO & FAST SPIN ECHO (SE/FSE)

THE WORKHORSE SEQUENCES FOR ROUTINE MRI

Spin Echo (SE) forms one echo per TR using a 180° refocusing pulse. Fast Spin Echo (FSE) collects many echoes per TR (echo train), making imaging much faster while maintaining T2 weighting.

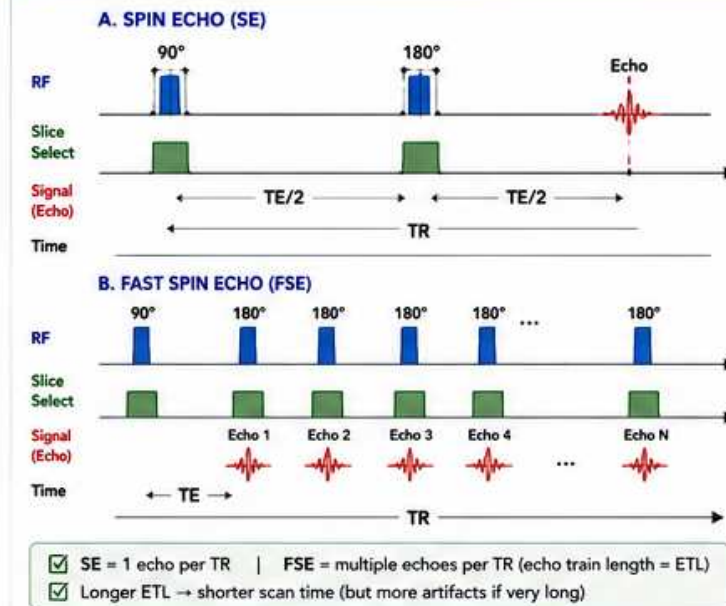


KEY CONCEPT

- SE = one echo per TR (slow but simple)
- FSE = many echoes per TR (fast, efficient)
- Both use 90° excitation + 180° refocusing
- Foundation for most clinical MRI

- ↑ Increase / Benefit
- ↓ Decrease / Trade-off
- No / Minimal Effect

1 HOW IT WORKS



2 WHY THE TECHNOLOGIST USES IT

- SE (Sometimes)**
- Very long TE for strong T2 weighting
 - When minimal blurring is critical
 - Research and special applications
 - Fewer echoes = less T2 decay blurring

- FSE (Most of the time)**
- Much faster than SE
 - High SNR and good contrast
 - Efficient use of time
 - Basis of T1, T2, PD, FLAIR, STIR, etc.

Bottom Line
 FSE provides the best balance of speed, SNR and image quality for most exams.

3 WHAT A NORMAL IMAGE GENERALLY LOOKS LIKE

Axial Brain Examples (1.5T)

T1-weighted (FSE)

- Fat: bright
- White matter: brighter than gray
- CSF: dark

T2-weighted (FSE)

- CSF: very bright
- Gray matter: bright
- White matter: darker than gray

PD-weighted (FSE)

- Subtle contrast
- Gray matter slightly brighter than white
- CSF: dark/gray

✓ Appearance varies with TR, TE and other parameters. These are general expectations, not absolute rules.

4 PARAMETERS THAT MATTER

Parameter	What It Does	Increase	Decrease
TR	More T1 recovery	More signal (less T1 weighting) Longer scan	More T1 weighting Shorter scan
TE	More T2 decay	More T2 weighting Lower signal (Longer scan)	Less T2 weighting Higher signal (Shorter scan)
ETL (Echo Train Length)	# of echoes per TR	Faster scan Higher SNR (up to a point) More T2 blurring More artifacts if very long	Slower scan Less T2 blurring Fewer artifacts
Slice Thickness	Coverage & SNR	Higher SNR More coverage	Lower SNR Less coverage
Bandwidth	Chemical shift & artifacts	Less chemical shift Shorter scan Lower SNR	More chemical shift Longer scan Higher SNR
NEX / Averages	SNR	Higher SNR Longer scan	Lower SNR Shorter scan

5 WHAT CAN GO WRONG (AND WHAT IT MAY INDICATE)

Problem / Artifact	What You See	Possible Causes (From an Engineering View)
T2 Blurring	Loss of sharpness in phase (direction of echo train)	• Long TE • Long ETL
Ghosting	Repeating ghosts along phase direction	• Motion • Flow • Eddy currents / gradient issues
Wrap / Aliasing	Anatomy appears on opposite side of image	• FOV too small • Phase oversampling low
Banding (FSE)	Dark/light bands across image	• B0 inhomogeneity • Long echo train (ETL)
Low Signal (Globally)	Overall low signal	• Coil problem • Wrong prescan • RF or system performance
Uneven Fat/Water (Shading)	Signal intensity varies across image	• B1 inhomogeneity • Coil loading / positioning

6 ENGINEER CHECKLIST (FOR SE / FSE)

- ✓ Is prescan center and shim adequate?
- ✓ Are TR, TE and ETL appropriate for the goal?
- ✓ Is FOV adequate to prevent wrap?
- ✓ Is phase direction optimal (A-P vs P-A)?
- ✓ Are bandwidth and NEX appropriate?
- ✓ Is the patient/coils positioned correctly?
- ✓ Is the coil functioning and well connected?
- ✓ Are gradients and RF systems performing within specs?
- ✓ Do artifacts match a known cause (motion, flow, B0, etc.)?
- ✓ Repeat test or sequence to confirm if needed.

Quick Parameter Impact (FSE) (All else constant)

- ↑ TR → ↑ Signal, less T1 weighting, longer scan
- ↓ TR → More T1 weighting, shorter scan
- ↑ TE → More T2 weighting, lower signal, longer scan
- ↑ ETL → Faster scan, higher SNR (up to a point), more T2 blurring, more artifacts
- ↓ ETL → Slower scan, less blurring, fewer artifacts

7 PRACTICAL TIPS

- Start with the protocol's recommended setup.
- Adjust one parameter at a time.
- Use the shortest scan time that still answers the clinical question.
- Always compare with prior images and other sequences.



REMEMBER

SE and FSE are the foundation of MRI. Most other sequences are built on the same principles.



KEY TAKEAWAY

SE = quality and simplicity.
 FSE = speed and efficiency.
 Choose the right balance for the right clinical question.